

Cambridge (CIE) IGCSE Chemistry
Paper 3: Theory (Set C)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 80

Instructions

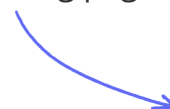
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) Substances can be classified as:

elements mixtures compounds

Elements can be divided into:

metals non-metals

Define each of the following terms.

i) element

[2]

ii) compound

[2]

iii) mixture

[1]

(5 marks)

(b) Classify each of the following as either an element, compound or mixture.

i) brass

[1]

ii) carbon dioxide

[1]

iii) copper

[1]

.....

.....

.....

(3 marks)

(c) Which physical property is used to distinguish between metals and non-metals?

It is possessed by all metals but by only one non-metal.

.....

(1 mark)

2 (a) The alkanes are a family of saturated hydrocarbons. Their reactions include combustion, cracking and substitution.

i) What is meant by the term *hydrocarbon*?

[1]

ii) What is meant by the term *saturated*?

[1]

.....
.....
(2 marks)

(b) i) What is the general formula for the homologous series of alkanes?

[1]

ii) Calculate the mass of one mole of an alkane with 14 carbon atoms.

[2]

.....
.....
.....
(3 marks)

(c) Cracking is used to obtain short-chain alkanes, alkenes and hydrogen from long-chain alkanes.

i) Give a use for each of the three products listed above.

short-chain alkanes

alkenes

hydrogen

[3]

ii) Write an equation for the cracking of decane, $C_{10}H_{22}$, which produces two different alkenes and hydrogen as the only products.

[1]

.....

.....

.....

.....

(4 marks)

3 (a) Use your copy of the Periodic Table to answer these questions.

Choose an element from the Periodic Table to match each description. You may give either the name or the symbol.

i) It is the most reactive metal.

[1]

ii) It is the only non-metal which is a liquid at r.t.p.

[1]

iii) This Group VII element is a solid at r.t.p.

[1]

iv) This element is in Group V and Period 4.

[1]

v) This unreactive gas is used to fill lamps.

[1]

.....

.....

.....

.....

.....

(5 marks)

(b) Predict the formula of each of the following compounds.

i) Germanium oxide

[1]

ii) Tellurium bromide

[1]

(2 marks)

(c) Give the formula of each of the following ions.

i) Strontium

[1]

ii) Fluoride

[1]

(2 marks)

4 (a) Universal indicator is added to an aqueous solution of sodium oxide.

What colour change is observed? from green to _____

Give a reason for your answer.

.....

.....

(2 marks)

(b) Acids have characteristic properties.

Hydrochloric acid reacts with magnesium carbonate.

Name the products of this reaction and give the observations.

.....

.....

.....

.....

(4 marks)

5 (a) This question is about iodine and compounds of iodine.

Use the kinetic particle model to describe the separation between the molecules and the type of motion of the molecules in:

Solid iodine:

Iodine gas:

.....

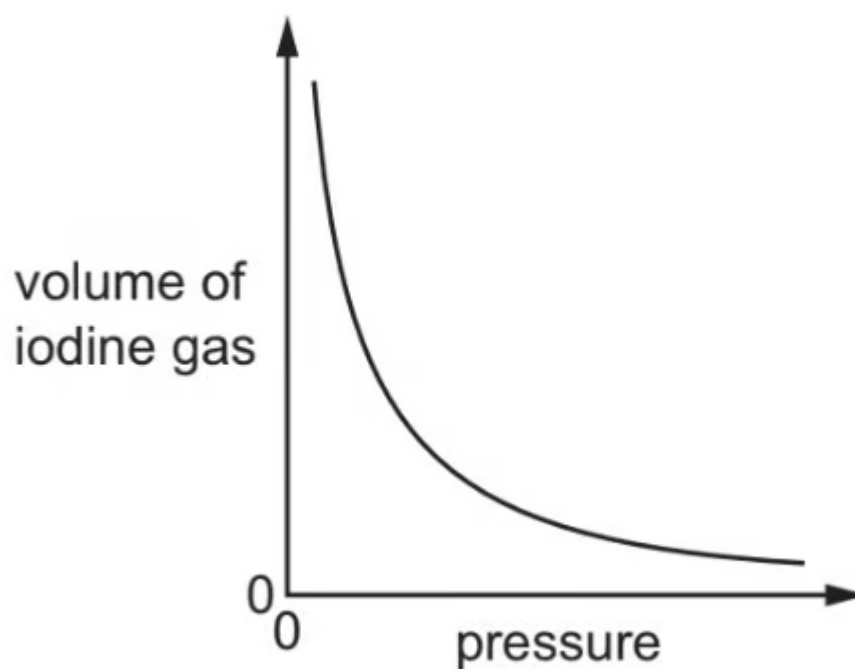
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(4 marks)

(b) The graph shows how the volume of iodine gas changes with pressure. The temperature is kept constant.



Describe how the volume of iodine gas changes with pressure.

.....

(1 mark)

(c) Sulfur dioxide melts at $-73\text{ }^{\circ}\text{C}$ and boils at $-10\text{ }^{\circ}\text{C}$.

What is the physical state of sulfur dioxide at $-20\text{ }^{\circ}\text{C}$?

Explain your answer.

(2 marks)

6 (a) Natural gas, which is mainly methane, is a fossil fuel.

i) What is meant by the term *fuel*?

[1]

ii) Name **two** other fossil fuels.

[2]

iii) Name a **solid** fuel which is not a fossil fuel.

[1]

(4 marks)

(b) Fossil fuels are formed by the anaerobic decomposition of organic matter. Anaerobic means in the absence of oxygen.

i) The organic matter contains hydrogen and carbon. Suggest the products that would be formed if the decomposition occurred in the presence of oxygen.

[2]

ii) What are the **two** main disadvantages in the widespread use of fossil fuels?

[2]

(4 marks)

- 7 (a)** The ore of aluminium is bauxite which is impure aluminium oxide. Alumina, pure aluminium oxide, is obtained from bauxite.

Aluminium is formed at the cathode when a molten mixture of alumina and cryolite, Na_3AlF_6 is electrolysed.

- i) Name two products formed at the anode in this electrolysis.

[2]

- ii) All the aluminium formed comes from the alumina not the cryolite. Suggest two reasons why the electrolyte must contain cryolite.

[2]

(4 marks)

(b) The purification of bauxite uses large amounts of sodium hydroxide.

i) Describe the chemistry of how sodium hydroxide is made from concentrated aqueous sodium chloride. The description must include at least one ionic equation.

[5]

ii) Making sodium hydroxide from sodium chloride produces two other chemicals. Name these two chemicals and state one use of each chemical.

chemical

use

chemical

use

[2]

.....

.....

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.....

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.....

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(7 marks)

- 8 (a)** A student investigates the progress of the reaction between dilute hydrochloric acid, HCl , and an excess of large pieces of marble, CaCO_3 , using the apparatus shown in Fig. 5.1.

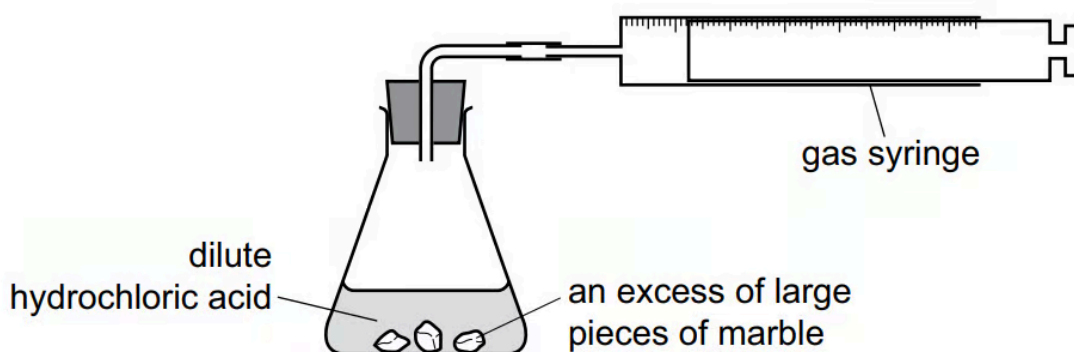


Fig. 5.1

A graph of the volume of gas produced against time is shown in Fig. 5.2.

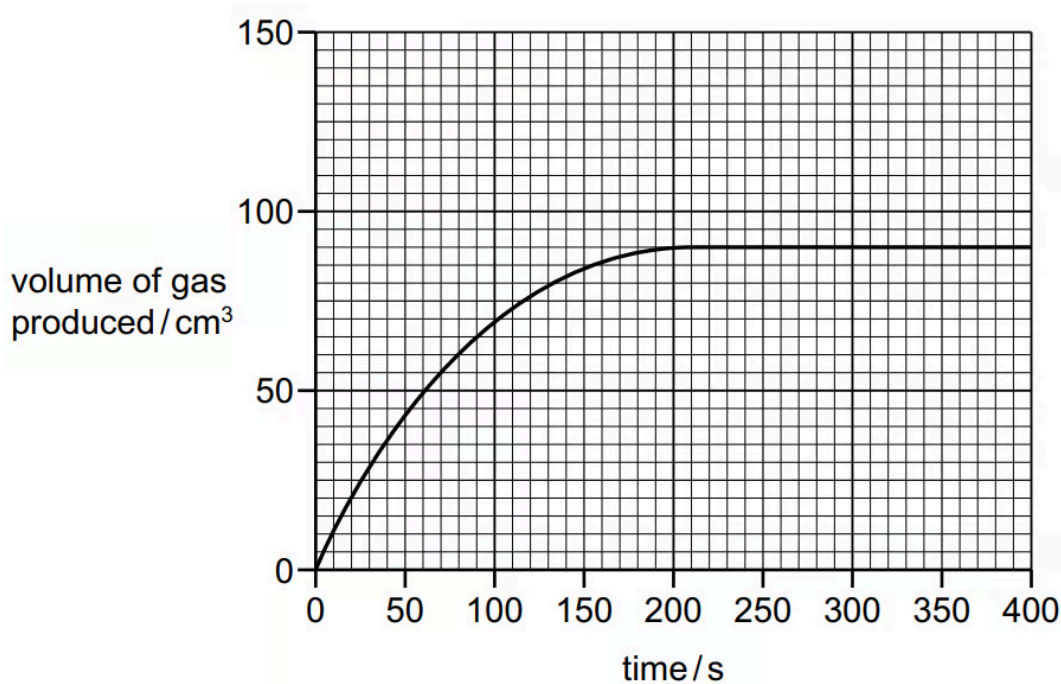


Fig. 5.2

State how the shape of the graph shows that the rate of reaction decreases as the reaction progresses.

(1 mark)

(b) Suggest why the rate of reaction decreases as the reaction progresses.

.....
(1 mark)

(c) Deduce the time at which the reaction finishes.

.....
(1 mark)

(d) The experiment is repeated using the same mass of smaller pieces of marble.

All other conditions are kept the same.

Draw a line on the grid in Fig. 5.2 to show the progress of the reaction using the smaller pieces of marble.

.....
.....
(2 marks)

- 9 (a)** Waste water like this needs to be treated before being released into the environment. Waste water may require removal of organic matter and harmful chemicals. The three main steps are sedimentation, filtration and chlorination.

Explain what happens in sedimentation.

(2 marks)

- (b)** Explain why chlorination is required.

(3 marks)

- (c)** **Table 5.1** shows some information about how the treated solid sludge was disposed of from one sewage treatment plant in 2000 and 2020.

Table 5.1

Mass of treated solid sludge in tonnes

Year	Used as fertiliser	Sent to landfill	Other	Total
2000	56	22	55	132
2020	152	2	39	193

Calculate the percentage of treated solid sludge that was sent to landfill in 2000.

Give your answer to 3 significant figures.

(3 marks)

- (d) The total mass of treated solid sludge increased from 2000 to 2020 and the proportions of how the sludge was disposed of changed.

Suggest **one** reason for the increase in the total mass of treated solid sludge and **two** reasons for the changes in the proportions of the disposal methods.

Use **Table 5.1** to help you answer.

(3 marks)

10 (a) This question is about metals and non-metals.

Compare the properties of metals and non-metals.

(4 marks)

(b) The Group I metals exhibit some properties that the majority of metals do not share.

Give **one** example of a property of Group I metals that is different from most other metals.

(1 mark)